
Revised Script for the Video Demonstration

Introduction (30 seconds)

[Start the video with your face visible or a slide showing your name, course, and assignment title.]

Hello, my name is **Amin**, and this is my submission for the **COM745BigData** assignment on **Crime Data Analysis Using Hive and Zeppelin**.

The purpose of this project is to analyze publicly available crime data to identify trends and patterns. The dataset I used contains information on violent and property crimes, categorized by state and county.

I used **Hadoop (HDFS)** for storage, **Hive** for SQL-based querying, and **Zeppelin** for creating visualizations.

Step 1: Data Upload Verification (1 minute)

[Switch to your screen showing the HDFS directory in the terminal.]

The dataset was uploaded earlier to Hadoop's HDFS for distributed storage. Let's verify that the dataset is in the correct directory.

Here's the command to list the contents of the directory /user/hadoop/dataset/:

hdfs dfs -ls /user/hadoop/dataset/

[Run the command and highlight the uploaded file in the output.]

As you can see, the file 2017report.csv is successfully uploaded to HDFS. This dataset will now be queried using Hive.

Step 2: Data Cleaning in Hive (1 minute)

[Explain how the dataset was cleaned using the pre-created table crime_report_cleaned. Show verification queries.]

The dataset was cleaned earlier to handle missing values and standardize column types. Let's verify the cleaned data:

SELECT * FROM crime_report_cleaned LIMIT 10;

[Show the query results and highlight a few rows.]

Here, you can see the cleaned dataset where missing values in columns like Murder_and_manslaughter and Rape have been replaced with 0.

[Show a query to confirm no null values remain in the dataset.]

SELECT

SUM(CASE WHEN Violentcrime IS NULL THEN 1 ELSE 0 END) AS null_violentcrime,

SUM(CASE WHEN Murder_and_manslaughter IS NULL THEN 1 ELSE 0 END) AS null_murder,

SUM(CASE WHEN Rape IS NULL THEN 1 ELSE 0 END) AS null_rape

FROM crime_report_cleaned;

[Explain the results.]

The output confirms that there are no null values in key columns, and the data is ready for analysis.

Step 3: Analysis in Hive (1.5 minutes)

[Run and explain the analysis query in Hive.]

1. **Total Crimes by State:**
2. SELECT State, SUM(Violentcrime) AS TotalViolentCrime
3. FROM crime_report_cleaned
4. GROUP BY State
5. ORDER BY TotalViolentCrime DESC;

This query calculates the total violent crimes for each state. States with higher totals, like [highlight a state], require targeted policy interventions.

[Summarize the results briefly and transition to visualizations.]

Step 4: Visualization in Zeppelin (1.5 minutes)

[Switch to Zeppelin and explain the visualizations created using the new query.]

1. **Query: Distribution of Crime Types by State**

SELECT

State,

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  SUM(Violentcrime * 2 + Propertycrime) AS CrimeSeverityIndex
FROM crime_report_cleaned
GROUP BY State
ORDER BY CrimeSeverityIndex DESC;
```

Explanation:

The Crime Severity Index highlights states with the most severe crime levels, accounting for both violent and property crimes.

Higher scores indicate more severe crimes, helping allocate law enforcement resources effectively.

Conclusion (30 seconds)

[Summarize the project and findings.]

In conclusion, this project provided insights into crime patterns using Hadoop, Hive, and Zeppelin:

- States with the highest crime rates were identified.
- The analysis highlighted variations in crime types across states, allowing for targeted interventions.
- Visualizations further revealed the overall distribution and trends of crimes.

These insights can help policymakers allocate resources effectively and improve public safety.

Thank you for watching my demonstration.
